Swayam Singh

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Education

University of Allahabad 2020 – 2024
B.Tech, Computer Science and Engineering CGPA: 9.56

Experience

Quansight
Software Engineer II · Full-time 07/2026 – Present

- Re-architected vLLM's multi-process serving stack into a single free-threaded (no-GIL) process, cutting inter-engine coordination overhead ∼13.3x and P99 inter-token latency from ∼540 ms to ∼112 ms.
- Contribute upstream to CPython, NumPy, PyTorch, and CuPy, ensuring compatibility, correctness, and performance across the scientific Python and AI stack.

Open Source Intern – Numerical Types & Precision (NumPy OSS) · Internship 07/2024 – 09/2024

- Led development of NumPy-QuadDType, a cross-platform 128-bit floating-point system for consistent quad precision beyond platform-dependent long double.
- Implemented full dtype ecosystem integration: casting, ufunc dispatch, scalar types, memory model, and serialization.
- Built multi-platform distribution and testing infrastructure handling compiler toolchains, SIMD availability, FMA support, and architecture-specific behavior.

Neocambrian AI
Member of Technical Staff – Reinforcement Learning

- Design and build RL environments for training and evaluating agentic models across cybersecurity, AI research, robot policy control, and vision reasoning, with verifiable, hacking-resistant rewards.
- Cybersecurity: agents find vulnerabilities and build working exploits by attacking simulated hardware (e.g., DMA), breaking cryptographic schemes, escaping sandboxes, and uncovering a planted cache-eviction bug.
- AI research, robotics & vision: agents that run ML experiments end-to-end under automated grading; physics-simulated manipulation with domain randomization; grounded, verifiable multimodal reasoning tasks.
- Building a Physical AI RSI (recursive self-improvement) framework for robot policy learning, where policies generate their own simulated experience to drive successive rounds of policy and reward refinement.

Microsoft Research 07/2024 – 06/2026
Research Fellow

- Built the world's first model and training pipeline for end-to-end autoformalization and verification of Rust programs, engineering the mid- and post-training recipe that turns informal specs into verifiable code.
- Developed RL algorithms exploiting gradient-optimization dynamics for representation efficiency, stability, and generalization; led large-scale multi-GPU distributed training.
- Trained an LLM on par with Gemini and GPT-4o for code generation and reasoning via post-pretraining and online/offline RL; developed SeleKT, an SFT method that preserves generalization while improving task accuracy.

NumPy (numpy-quaddtype) 12/2024 – Present
Maintainer

- Review, merge, and evaluate core PRs affecting dtype architecture, numerical semantics, and cross-platform behavior.
- Contribute to core NumPy with a focus on dtype infrastructure, numerical correctness, and long-term stability.
- Work with contributors across architectures on portability, performance, and ABI concerns.

dataX.ai 05/2022 – 10/2022
Machine Learning Engineer Intern

- Built deep-learning models for vision and language; created an ONNX conversion API with Triton deployment, cutting VM load by 12%.
- Wrote custom CUDA kernels for 3D medical-scan processing, achieving a 2x speedup in segmentation over existing GPU solutions.

Publications

- **NextCoder: Robust Adaptation of Code LMs to Diverse Code Edits** (ICML 2025; ICLR 2025 DL4Code). **Swayam Singh**, Tushar Aggarwal, et al. A synthetic-data generation pipeline and SeleKT (Selective Knowledge Transfer), a new fine-tuning algorithm that makes code LLMs robust to diverse, real-world code edits. Our 32B model matches Gemini and beats GPT-4o on harder tasks like aider-polyglot, achieved purely via SFT, with SeleKT substantially reducing catastrophic forgetting compared to other methods.

- **Narrow Transformer: StarCoder-Based Java-LM For Desktop** (arXiv:2407.03941, 2024).
  Kamalkumar Rathinasamy, ..., **Swayam Singh**, et al. A compact, Java-specialized code language model designed to run efficiently on desktop hardware.
- **OctoPack: Instruction Tuning Code Large Language Models** (ICLR 2024 Spotlight; NeurIPS 2023 Instruction Workshop).
  Niklas Muennighoff, ..., **Swayam Singh**, et al. Instruction tuning of code models using natural-language Git commits (CommitPack / CommitPackFT).
- **StarCoder: May the Source Be With You!** (TMLR 2023).
  Raymond Li, ..., **Swayam Singh**, et al. (BigCode). A 15.5B-parameter open code LLM trained on 1T tokens of permissively licensed code.

Projects

- **Virtual Clothing Assistant (PyTorch)**: An end-to-end virtual try-on system: ResNet101 and UNet for garment/body segmentation, OpenPose for pose estimation, and a PyTorch VITON pipeline for realistic garment warping. Reaches an SSIM of 0.895 and has 500+ GitHub stars.
- **Numpy-QuadDType (C, C++, Python)**: A cross-platform 128-bit (quadruple-precision) floating-point dtype for NumPy with 100k+ downloads. A portable alternative to long double with full casting, ufunc dispatch, scalar types, and serialization that behaves consistently across compilers and architectures.
- **Bare-Bones Inference (CUDA, C++)**: A minimal inference stack with engine-grade features such as batching and scheduling, but serving a single fused megakernel instead of a conventional multi-kernel pipeline. Built DeepSeek-V2 (236B)'s block-scaled FP8-quantized, multi-GPU megakernel for NVIDIA B200 (sm_100) GPUs that fuses all computation and TP/EP communication into one kernel, handling cross-rank exchange via `multimem` PTX multicast writes (directly writing into peer ranks' memory at once). Achieves 13.11 ms/token decode latency, beating vLLM's optimized serving latency at B=1 by 2x and eager-mode by >10x.
- **QBLAS (C, C++)**: A high-performance BLAS library for IEEE-754 binary128 (quadruple) precision, implementing optimized linear-algebra kernels that bring quad-precision numerics to workloads unsupported by standard double-precision libraries.
- **cpp-verify (C++, LLVM, SMT)**: Extends C++ with first-class formal-verification constructs (pre/post-conditions and invariants), lowering specifications through an LLVM-based pipeline and discharging the resulting proof obligations to SMT solvers.
- **MIRA – Multimodal Image Reconstruction with Attention (PyTorch)**: A transformer-based architecture for single-view text/image-to-3D reconstruction using ViT encoders and triplane decoders with cross-attention. Generates 3D mesh and video in under 10s on an A100, with SDXL-integrated diffusion for end-to-end scene synthesis.

Honors and Awards

- **2024** — **Kaggle Competition Expert**: Bronze medal (top 7%) in the UBC-OCEAN competition; top 3% in the *30 Days of ML* challenge.
- **2024** — Invited to **Google Research Week**, Google Research's gathering of AI researchers (keynote by Jeff Dean; sessions on differential privacy, responsible AI, and more).
- **2024** — **OctoPack** accepted as a **Spotlight (top 5%)** at ICLR 2024.
- **2023** — Selected for the **Amazon ML Summer School 2023**.
- **2023** — **Clothes Virtual Try-On** crossed 500+ GitHub stars.

Invited Talks

- **MAMBA: Zero to Hero**, invited talk on State Space Models at Cohere for AI.
- **Provably-Correct Code and Efficient Sparse Training of LLMs**, internal research talk at Microsoft Research.
- **Foundations of Machine Learning**, a GDG On-Campus session on the ML landscape, tooling ecosystem, and emerging directions.
- **From Deep Learning to Large Language Models**, a talk on the foundations of modern AI: deep learning, generative models, and LLMs.

Skills Summary

- **Languages**: C, C++, CUDA, CuTeDSL, Python
- **ML & Training**: LLM post-training, reinforcement learning (online/offline RL), supervised fine-tuning, multi-GPU distributed training, code-generation models
- **Systems & GPU**: CUDA kernels and megakernels, high-performance numerical computing, IEEE-754 quad precision, SIMD/FMA, performance optimization
- **Tools & Frameworks**: PyTorch, NumPy (C-API internals), LLVM, SMT solvers
- **Open Source**: Core NumPy maintenance, cross-platform C-extension packaging, code review